

Week 4 object memory budget study

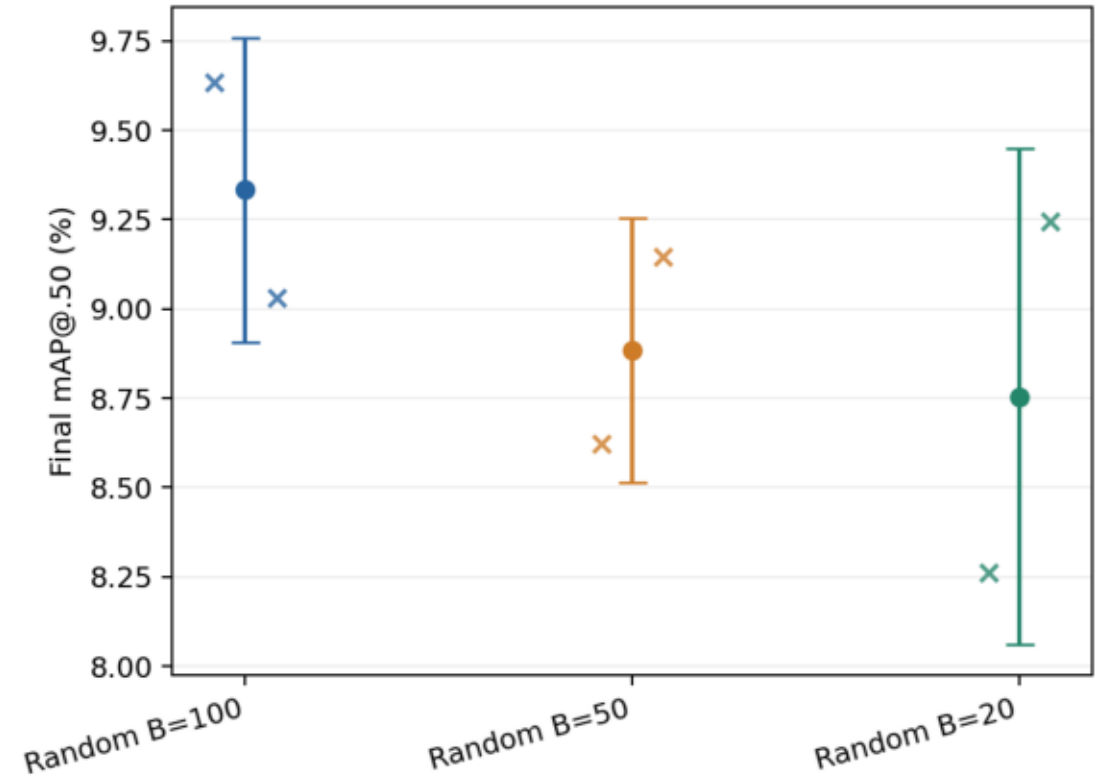
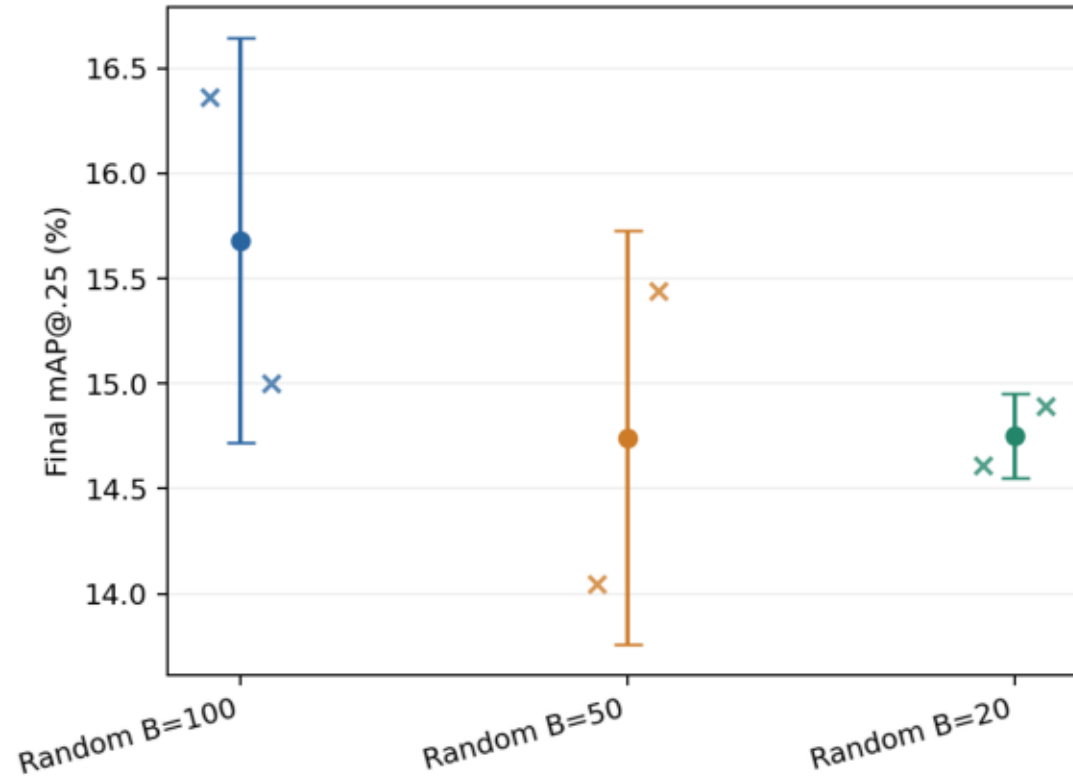
- Ten stages · 40 SUN RGB-D classes · two full-training seeds
- 3,979 → 800 stored objects: 79.89% fewer
- Final AP25: 15.6815% → 14.7520% (−0.9295 pp)
- The smaller bank does not meet the provisional 0.5 pp loss tolerance.

Protocol and comparison scope

- Ten frequency-ordered stages, four new classes per stage.
- $B=100, 50$ and 20 ; seeds 200 and 201 train from scratch.
- Fixed schedule, pseudo supervision and configured insertion dose.
- Week 3 changed replay probability; Week 4 changes stored capacity.
- Reference budget and split matched; no verified S10 object score.

Final accuracy across budgets

Ten stages, two full-training seeds | mean and sample SD; crosses are seeds

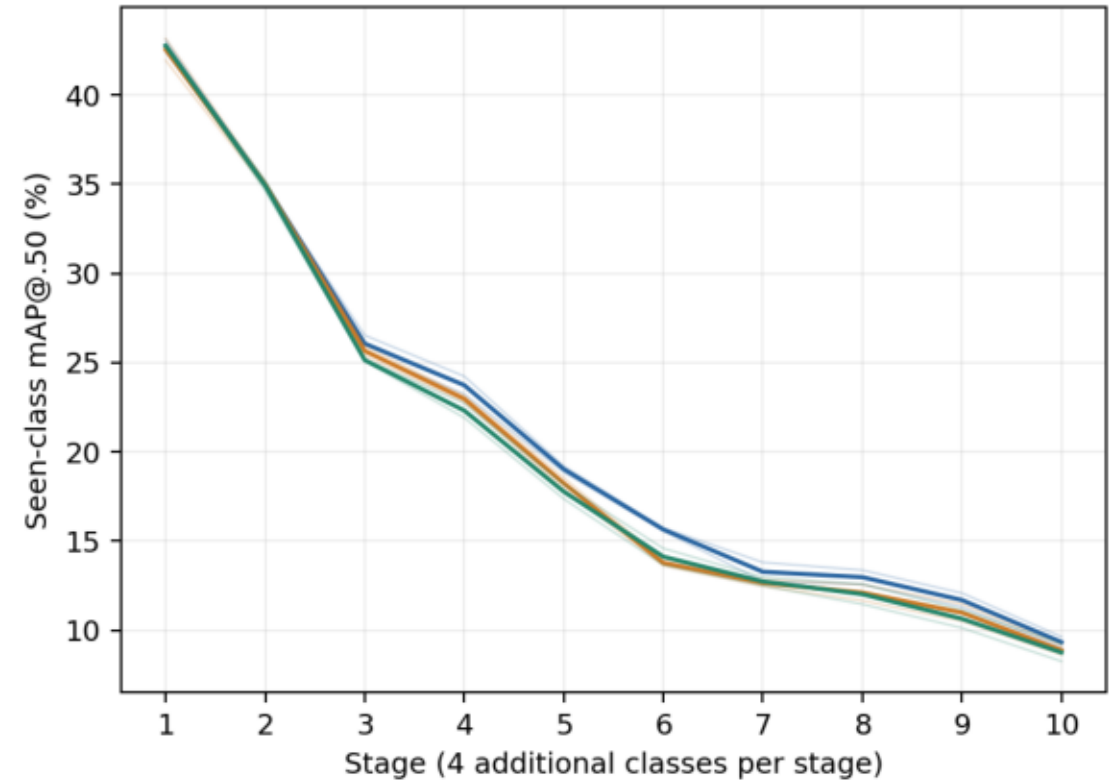
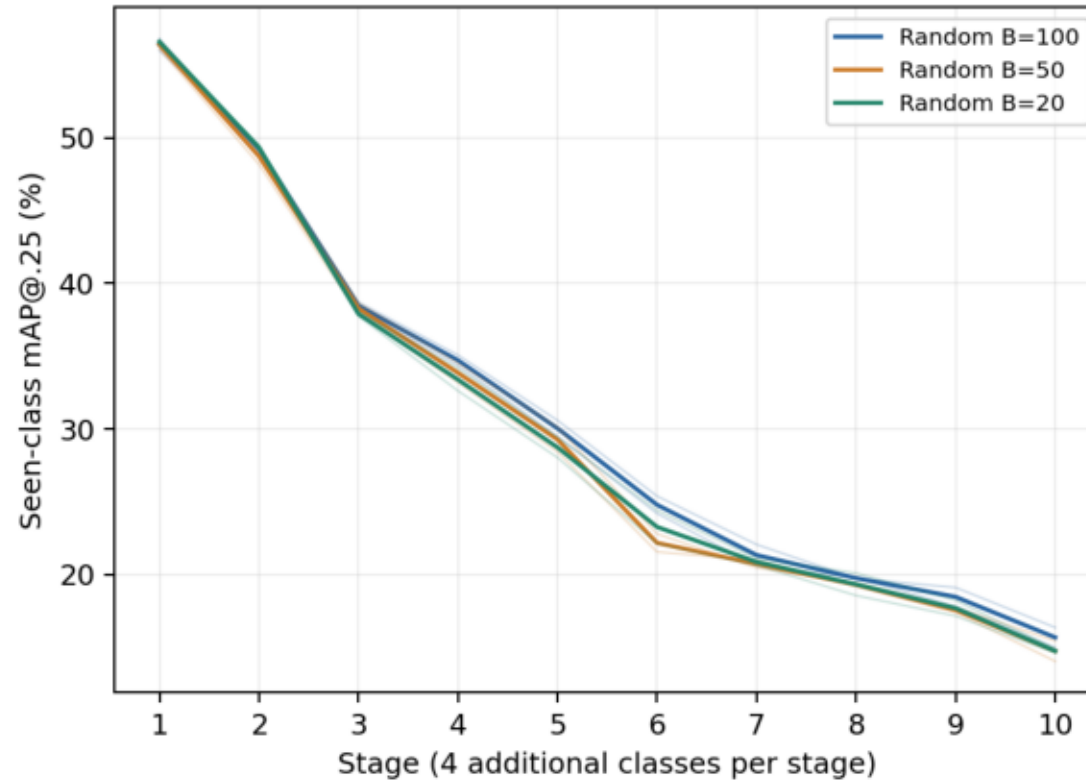


B=20 paired AP25 changes: -1.754 and -0.105 pp.

Neither reduced budget passes the provisional 0.5 pp mean tolerance.

Stage trajectory and forgetting

Mean stage trajectory; faint lines are individual seeds

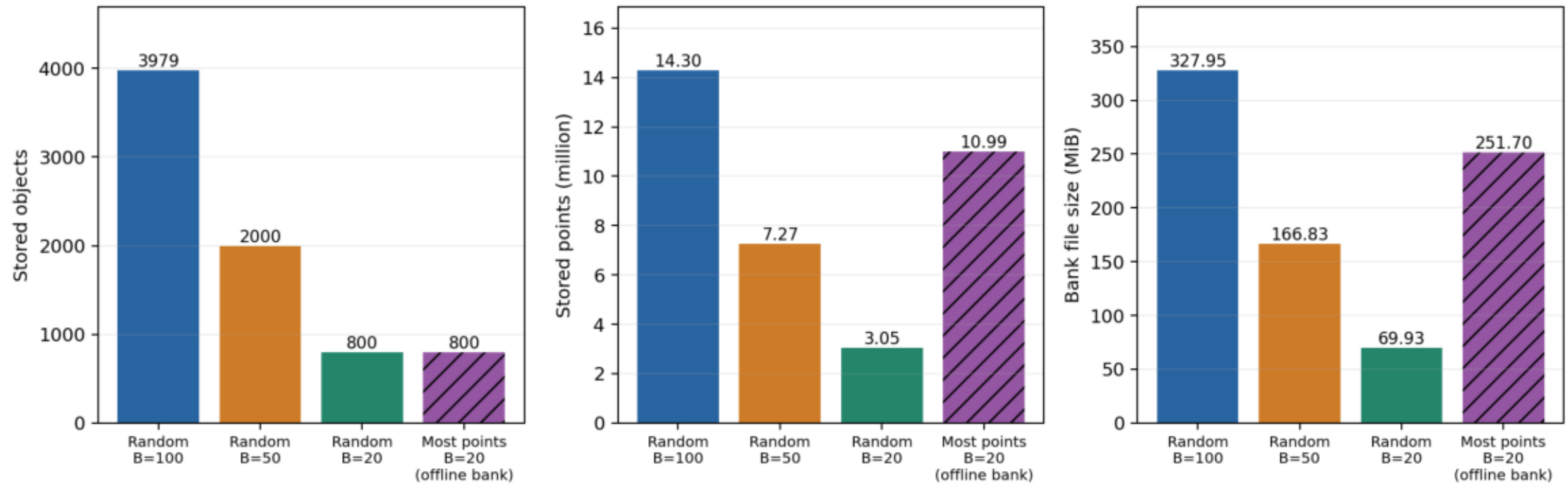


Seen-class scope grows at each stage.

Raw mAP decline alone is not a forgetting measure.

Equal object counts have different byte costs

Same object count does not imply the same point or byte budget
Random banks: mean of two trained runs; dense bank: audited GT construction only

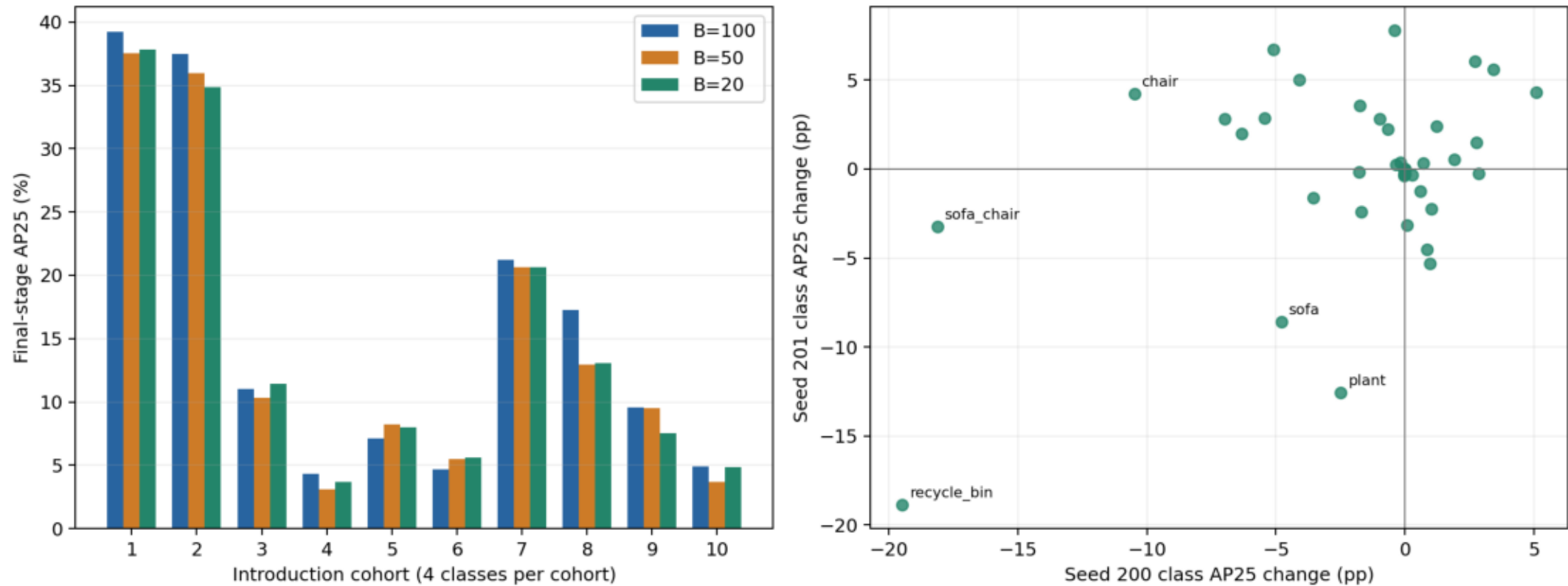


Dense B=20 bank: 251.70 MiB; 3.60× random B=20.

Dense bank is offline GT construction; no full S10 detector score.

Class effects and seed variability

Budget effects vary across class cohorts and seeds
Left: two-seed means; right: B=20 minus B=100 per class; descriptive evidence



recycle_bin contributes -0.4793 pp to the total -0.9295 pp AP25 change.

Descriptive decomposition; the cause of the loss is not established.

Selection repair and bounded diagnostic

- Full selector runs failed at stage-1 bank population; integration path repaired.
- Same stage-1 checkpoint within each pair; stage 2 only, eight seen classes.
- Largest minus random AP25: +0.870, +1.080 pp.
- Mean early-stage change: +0.9750 pp.
- Pseudo labels differ too; the gain is not isolated to the selector.
- This does not establish final ten-stage performance.

What we can conclude

- Random $B=20$ gives large storage savings at about 0.93 pp mean AP25 cost.
- Retention within 0.5 pp has not been demonstrated.
- Two seeds do not establish statistical equivalence.
- Next: finish S10 selection with shared pseudo labels; add a full seed.
- This week closes with measured results and the missing comparison stated.